



FRAGRANCE TABLE

Chemistry of Oo

Olfactive Taxonomy Specification

Version	1.0
Date	2026
Status	Active — open for community contribution
Maintainer	Oo La Lab
License	Creative Commons Attribution 4.0 International (CC BY 4.0)
Scope	27 olfactive accords · periodic table format · drill-down ingredient layer

Abstract

The Chemistry of Oo Olfactive Taxonomy is an open classification system for fragrance accords. It defines 27 discrete olfactive accords — organised by molecular weight from lightest to heaviest — as the primary reference layer for describing, comparing, and encoding fragrance compositions.

The taxonomy is designed to be machine-readable, structurally rigorous, and practically applicable by perfumers, educators, researchers, and developers. Each accord acts as a coordinate in olfactive space. A fragrance formula is described by its profile across the 27 accords — enabling similarity search, gap analysis, and AI training on a consistent vocabulary.

The system is published as an open specification under CC BY 4.0. Anyone may use, implement, or build upon it. Oo La Lab maintains stewardship over the schema, versioning, and contribution process.

Design principles

Molecular weight axis

Accords are ordered from lightest to heaviest molecular weight — left to right, top to bottom — mirroring the physical unfolding of a fragrance on skin. This is not a stylistic choice but a structural one: the taxonomy's spatial layout encodes chemical behaviour.

Top notes (lightest, most volatile) sit in the upper rows. Heart notes occupy the middle row. Base notes (heaviest, most persistent) occupy the lower rows. This means that reading the table from top-left to bottom-right follows the temporal arc of wearing a fragrance.

Three tiers, four symbols

Each accord carries a tier symbol that communicates its volatility class:

- ○ Circle — top notes. Highest volatility, fastest diffusion, lightest molecular weight.
- △ Triangle — heart notes. Mid volatility, structural core of the composition.
- □ Square — base notes. Lower volatility, long dry-down, structural fixatives.
- ◇ Diamond — heaviest base accords. Animalic, resinous, and skin-close fixatives.
- ⬠ Pentagon — historical reference accords (Fu, Cy). Cross-tier constructs, not volatility-classified.

Fu and Cy — historical reference accords

Fougère (Fu) and Chypre (Cy) occupy a distinct position in the table. They sit apart from the main grid because they are not single-note accords but historical accord families — each defined by a specific combination of multiple notes that together produce a recognisable olfactive signature.

Their separation is intentional and meaningful: they function as compositional reference points against which other formulas can be located, rather than as building blocks in the same sense as the 25 main accords. A formula described as 'Cy-adjacent' is understood to share the dry, earthy-citrus tension of the Chypre family even if it deviates from the strict oakmoss-labdanum-bergamot triad.

The gap cells

Blank cells in the table are intentional structural whitespace — analogous to the gaps in the periodic table of elements. They preserve the spatial logic of the molecular weight axis and may be occupied by future accords if the taxonomy is extended through the RFC process. They are not errors or placeholders.

Drill-down: accords and ingredients

Each accord is the top level of a two-tier hierarchy. Below each accord sits a set of constituent ingredients — raw materials, aroma chemicals, and naturals — that express that accord's character. For example, Wo (Woods) drills down into cedarwood, sandalwood, vetiver, patchouli, guaiac wood, and others.

This structure means the taxonomy operates at two levels of resolution: accord-level for broad classification and comparison, ingredient-level for formulation detail and sourcing. Both levels share the same namespace and coding convention.

Palette layer

Palettes are curated subsets of accords assembled for a specific use context — occasion, season, demographic, mood, or application. A palette does not add new accords; it selects and weights from the existing 27. Palettes are defined in a separate layer of the specification and can be contributed by the community under the same CC BY 4.0 terms.

The 27 accords

The table below defines all 27 accords. For each accord: the two-letter code, full name, olfactive character, and representative example ingredients at the ingredient drill-down level. Codes are case-sensitive; the first letter is always uppercase.

Code	Accord	Olfactive character	Example ingredients
Historical reference accords — sit apart from the main table; cross-tier constructs			
Fu	Fougère	Herbaceous, mossy, lavender-led accord originating in 19th-century perfumery; reference family for masculine aromatic compositions	Lavender, oakmoss, coumarin, geranium
Cy	Chypre	Oakmoss, labdanum, bergamot triad defining a dry, earthy-citrus tension; foundational accord of modern perfumery	Oakmoss, labdanum, bergamot, cistus
Top notes — lightest molecular weight · symbol: ○			
Ci	Citrus	Bright, volatile, high-energy opening; zesty and effervescent; dissipates rapidly	Bergamot, lemon, grapefruit, neroli
Al	Aldehyde	Synthetic, waxy, soapy radiance; amplifies other notes; hallmark of classic haute parfumerie	C11, C12, fatty aldehydes
Aq	Aquatic	Marine, ozonic, clean water effect; evokes sea air and fresh surfaces	Calone, dihydromyrcenol, ambergris synthetics

Code	Accord	Olfactive character	Example ingredients
Oz	Ozone	Airy, electric, pre-storm atmosphere; abstract freshness with minimal warmth	Iso E Super light fraction, Helional, Ozocyclene
Fr	Fruity	Ripe, juicy, synthetic or natural fruit character excluding peach; broad modern application	Ethyl maltol, various esters, cassis absolute
Pe	Peach	Soft, velvety lactonic fruit; bridges fruity top notes and floral heart; distinctly tactile	Gamma-undecalactone, peach aldehyde, apricot esters
Te	Tea	Dry, tannin-edged, quietly green; hovers between top and heart; cerebral and restrained	Tea absolute, indole (trace), green tea CO2
Gr	Green	Cut grass, tomato leaf, galbanum, violet leaf; vegetal sharpness; bridges top and heart	Galbanum, violet leaf absolute, cis-3-hexenol
Ar	Aromatic	Herbal, camphoraceous, culinary-adjacent; lavender, rosemary, sage; structured and dry	Lavender, rosemary, clary sage, basil
Heart notes — mid molecular weight · symbol: Δ			
Fl	Floral	Broad floral character not assigned to a specific bloom; abstract flower, often powdery	Rose oxide, hedione, phenylethyl alcohol
Vi	Violet	Powdery-green, ionone-driven floral; simultaneously fresh and soft; bridges top-note greenness with heart floral depth	Ionone alpha, ionone beta, violet leaf absolute, orris (light fraction)
Mg	Muguet	Lily of the valley; clean, dewy, transparent white floral; the lightest heart accord	Lilial, Lylal, hydroxycitronellal, Bourgeonal
Wf	White Flowers	Indolic, heady, narcotic florals; tuberose, gardenia, narcissus; rich and complex	Tuberose absolute, indole, methyl anthranilate
Ja	Jasmine	Warm, slightly animalic floral; indole-forward; the most complex natural flower accord	Jasmine absolute, cis-jasmone, methyl jasmonate
Ro	Rose	Geraniol-dominant, structured floral; ranges from dewy fresh to deep damascone richness	Rose absolute, geraniol, beta-damascone, citronellol
Os	Orris	Powdery, violet-adjacent, rooty depth; bridges floral heart to base; distinctly tactile	Orris butter, irones, isomethyl ionone

Code	Accord	Olfactive character	Example ingredients
Sp	Spicy	Warm spice notes — pepper, cardamom, cinnamon, clove; bridges heart to base	Eugenol, cinnamic aldehyde, cardamom CO2, black pepper
Base notes — heavy molecular weight · symbol: $\square$ / $\diamond$			
Le	Leather	Birch tar, quinoline, suede; dry, smoky, animalic base; distinctly material character	Birch tar, quinoline, isobutyl quinoline, Norlimbanol
Or	Oriental	Resinous, balsamic, warm spice construction; the richest structural base family	Benzyl benzoate, Peru balsam, opoponax, labdanum
Va	Vanilla	Sweet, lactonic, gourmand-adjacent; vanillin and coumarin-led; warm and enveloping	Vanillin, ethyl vanillin, coumarin, heliotropin
Go	Gourmande	Edible, confectionery, café character; distinctly modern base category; sweet-dry tension	Ethyl maltol, praline accord, coffee absolute, caramel
Am	Amber	Warm, resinous, luminous; labdanum-benzoin-styrax triad; the canonical base accord	Labdanum absolute, benzoin resinoid, styrax, ambroxan
Wo	Woods	Dry, structural, multi-species; cedarwood through patchouli; backbone of base construction	Cedarwood, sandalwood, vetiver, patchouli, guaiac wood
Od	Oud	Barnyard-leathery-woody; agarwood and its synthetics; among the heaviest naturals	Oud oil, oud synthetics (Agarwood), iso E super (deep)
Mu	Musk	Skin-close, diffusive, fixative base; ranges from clean laundry to animalic depth	Habanolide, Galaxolide, Exaltolide, Muskenone

* Ingredient examples are illustrative, not exhaustive. Full ingredient drill-down lists are maintained in the companion dataset.

Classification guidelines

Assigning accords to a formula

A fragrance formula is classified by identifying which accords are perceptibly present and weighting them by intensity and dominance. Classification is accord-level, not ingredient-level —

the question is not 'does this formula contain jasmine absolute?' but 'does this formula express the Ja (Jasmine) accord character?'

The recommended workflow is:

1. Evaluate the formula's opening. Identify which top-note accords (○) are present and their relative weight.
2. Evaluate the heart. Identify which △ accords form the compositional core.
3. Evaluate the dry-down. Identify which □ and ◇ accords anchor and fix the composition.
4. Assess Fu/Cy relevance. If the overall construction references either historical family, note this separately.
5. Express the profile as a weighted accord vector — primary (dominant), secondary (supporting), and tertiary (trace) accords.

Accord weighting

Each accord in a formula's profile carries one of three weights:

- Primary (P) — the accord is dominant and immediately perceptible. Maximum 3 primary accords per formula.
- Secondary (S) — the accord is clearly present but supporting. No hard limit.
- Tertiary (T) — the accord is detectable in context, contributing nuance but not character.

Example profile notation: Ro(P) · Wf(P) · Am(S) · Mu(S) · Al(T)

What counts as an accord

An accord must be perceptibly present on the skin or smelling strip — not merely present as an ingredient in the formula. A trace of vanilla that registers only on cold sniff and disappears on warmth would not qualify as Va(P) or even Va(S).

Accords may overlap and interact. A formula can legitimately carry both Ro (Rose) and FI (Floral) if the rose character is specific and distinct AND a broader, less-specific floral note is also present.

Versioning and governance

Version history

Version	Date	Notes
1.0	2026	Initial public specification. 27 accords. Tier symbol system. Classification guidelines. Palette layer defined.

RFC process — proposing changes

Changes to the taxonomy — new accords, accord redefinitions, tier reassignments, or changes to classification guidelines — are proposed via a public Request for Comment (RFC) process.

6. Open an RFC issue in the public repository describing the proposed change and rationale.
7. A 30-day public comment period opens. Anyone may respond.
8. The maintainer reviews comments and either accepts, modifies, or declines the RFC.
9. Accepted changes are incorporated in the next version release with full changelog documentation.

License

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The companion dataset (classified fragrance corpus) is published separately under CC BY 4.0. Individual formula contributions retain attribution to their contributors.

How to contribute

Contributions are welcome from perfumers, students, researchers, and developers. The taxonomy is a community resource and improves with wider use and scrutiny.

Classify a fragrance: Apply the classification guidelines to a formula or finished fragrance and submit the accord profile via the public repository.

Propose an ingredient: Suggest additions to the drill-down ingredient list for any accord via a pull request.

Open an RFC: If you believe an accord definition should be refined or a new accord added, open an RFC as described above.

Translate: The specification welcomes translations into other languages. Translations are clearly marked as community translations and do not alter the canonical English text.

Cite: Academic and professional use should cite this document as: Chemistry of Oo Olfactive Taxonomy Specification, v1.0 (2026). chemistry-of-oo.com

Acknowledgements

The Chemistry of Oo taxonomy was developed through hands-on fragrance education, workshop practice, and iterative refinement with students and practitioners across the global fragrance community. The periodic table format was chosen deliberately — both as a reference to the chemical foundations of perfumery and as a visual language that makes the taxonomy immediately navigable to anyone who encounters it.

The system owes its existence to the perfumers, students, and enthusiasts who have used it, questioned it, and helped refine its edges. Their practice is encoded in every accord definition.

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